Spectrally Sparse Nonparametric Regression via Elastic Net Regularized Smoothers

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Multiple and Generalized Nonparametric Regression

MGNR frameworks are popular for discovery and prediction

- GAM = generalized additive model (Wood, 2017)
- SSANOVA = smoothing spline analysis of variance (Gu, 2013)

Penalized likelihood estimation used to estimate unknown functions

$$\text{PLS} = -\frac{1}{n}\text{ Log-Likelihood} + \lambda \text{ Penalty}$$

where n is sample size and $\lambda > 0$ is smoothing parameter.

Problem: tuning and model selection can be computationally costly

Model Selection and Smoothing

Past methods for selection and smoothing:

- COSSO (Lin & Zhang, 2006)
- SpAM (Ravikumar et al., 2008)
- Sparse Penalty (Meier et al., 2009)
- GAMSEL (Chouldechova & Hastie, 2015)
- Heuristics and IC (Marra & Wood, 2011; Wood et al., 2016)

Most of these methods use some variant of. . .

- LASSO (Tibshirani, 1996)
- group LASSO (Yuan & Lin, 2006)

Overview of Proposed Idea

Goal: Robust smoothing and prediction for misspecified models

Does not assume there is a “correct” additive model generating data

Kernel eigenvector smoothing and selection operator (kesso)

- Reparameterize smoothing spline problem
- Apply elastic net penalty to kernel eigenvectors
- Efficient tuning using GCV criterion

Result: kesso finds truth (simulation) and predicts well (real data)

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Polynomial Smoothing Spline Problem

Given n independent observations $\{(x_i, y_i)\}_{i=1}^n$, assume the model

$$y_i = \eta(x_i) + \epsilon_i$$

where η is an unknown smooth function of $x_i \in [0,1]$ and $\epsilon_i \overset{\text{iid}}{\sim} (0, \sigma^2)$.

To estimate η, consider the penalized least squares problem

$$\eta_\lambda = \min_{\eta \in \mathcal{H}_m} \left\{ \frac{1}{n} \sum_{i=1}^n w_i \left(y_i - \eta(x_i)\right)^2 + \lambda \int_0^1 |\eta^{(m)}(x)|^2 dx \right\}$$

where $w_i > 0$, $\eta^{(m)} = \frac{d^m \eta(x)}{dx^m}$, and $\mathcal{H}_m = \{\eta : \int_0^1 |\eta^{(m)}(x)|^2 dx < \infty\}$

Representer Theorem

The representer theorem (Kimeldorf & Wahba, 1971) reveals that

$$\eta_\lambda(x) = \sum_{v=0}^{m-1} \beta_v \phi_v(x) + \sum_{i=1}^{r} \gamma_i \rho(x, k_i)$$

where

- $\{\phi_v\}_{v=0}^{m-1}$ span the null space $\mathcal{H}_m^0 = \{\eta : \int_0^1 |\eta^{(m)}(x)|^2 dx = 0\}$
- ρ is the reproducing kernel of the contrast space $\mathcal{H}_m^1 = \mathcal{H}_m \ominus \mathcal{H}_m^0$
- $0 \leq k_1 < \cdots < k_r \leq 1$ denote selected knots (RT uses x_i as knots)
- $\boldsymbol{\beta} = (\beta_0, \ldots, \beta_{m-1})'$ and $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_r)'$ are coefficients

Can use a subset of x_i as knots (Kim & Gu, 2004; Ma et al., 2015).

Penalized Least Squares Revisited

Using the representer theorem, the PLS problem has the form

$$\frac{1}{n}\|\mathbf{W}^{1/2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\boldsymbol{\gamma})\|^2 + \lambda\boldsymbol{\gamma}'\mathbf{Q}\boldsymbol{\gamma} \tag{1}$$

where

- $\mathbf{W} = \text{diag}(w_1, \ldots, w_n)$ are the weights (typically $w_i = 1\ \forall i$)
- $\mathbf{y} = (y_1, \ldots, y_n)'$ is the response vector ($y_i \in \mathbb{R}\ \forall i$)
- $\mathbf{X} = [\phi_u(x_i)]$ is the $n \times m$ null space basis function matrix
- $\mathbf{Z} = [\rho(x_i, k_v)]$ is the $n \times r$ contrast space basis function matrix
- $\mathbf{Q} = [\rho(k_j, k_v)]$ is the penalty matrix

Coefficient Estimates

Given λ the coefficient estimates can be written as

$$\begin{pmatrix} \hat{\boldsymbol{\beta}}_\lambda \\ \hat{\boldsymbol{\gamma}}_\lambda \end{pmatrix} = \begin{pmatrix} \mathbf{X}_w'\mathbf{X}_w & \mathbf{X}_w'\mathbf{Z}_w \\ \mathbf{Z}_w'\mathbf{X}_w & \mathbf{Z}_w'\mathbf{Z}_w + n\lambda\mathbf{Q} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{X}_w' \\ \mathbf{Z}_w' \end{pmatrix} \mathbf{y}_w \tag{2}$$

where

- $\mathbf{X}_w = \mathbf{W}^{1/2}\mathbf{X}$ and $\mathbf{Z}_w = \mathbf{W}^{1/2}\mathbf{Z}$ are the weighted design matrices
- $\mathbf{y}_w = \mathbf{W}^{1/2}\mathbf{y}$ is the weighted response vector

The form of the inverse of a block matrix implies that

$$\hat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}_w'\mathbf{X}_w)^{-1}\mathbf{X}_w'(\mathbf{y}_w - \mathbf{Z}_w\hat{\boldsymbol{\gamma}}_\lambda)$$
$$\hat{\boldsymbol{\gamma}}_\lambda = (\mathbf{P}_w'\mathbf{P}_w + n\lambda\mathbf{Q})^{-1}\mathbf{P}_w'\mathbf{y}_w$$

where $\mathbf{P}_w = (\mathbf{I} - \mathbf{H}_w)\mathbf{Z}_w$ and $\mathbf{H}_w = \mathbf{X}_w(\mathbf{X}_w'\mathbf{X}_w)^{-1}\mathbf{X}_w'$

Fitted Values and Smoothing Matrix

Given the optimal coefficients, the fitted values have the form

$$\hat{\mathbf{y}}_\lambda = \mathbf{X}\hat{\boldsymbol{\beta}}_\lambda + \mathbf{Z}\hat{\boldsymbol{\gamma}}_\lambda = \mathbf{W}^{-1/2}\mathbf{S}_\lambda \mathbf{y}_w \tag{3}$$

where

$$\mathbf{S}_\lambda = \begin{pmatrix} \mathbf{X}_w & \mathbf{Z}_w \end{pmatrix} \begin{pmatrix} \mathbf{X}_w'\mathbf{X}_w & \mathbf{X}_w'\mathbf{Z}_w \\ \mathbf{Z}_w'\mathbf{X}_w & \mathbf{Z}_w'\mathbf{Z}_w + n\lambda\mathbf{Q} \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{X}_w' \\ \mathbf{Z}_w' \end{pmatrix}$$

is the smoothing matrix.

Note that $\mathbf{S}_\lambda = \mathbf{H}_w + \mathbf{S}_\lambda^1$, where $\mathbf{S}_\lambda^1 = \mathbf{P}_w(\mathbf{P}_w'\mathbf{P}_w + n\lambda\mathbf{Q})^{-1}\mathbf{P}_w'$

Smoothing Parameter Selection

The smoothing parameter λ is often selected by minimizing the generalized cross-validation (GCV) criterion (Craven & Wahba, 1979)

$$\mathrm{GCV}(\lambda) = \frac{\frac{1}{n}\|(\mathbf{I} - \mathbf{S}_\lambda)\mathbf{y}_w\|^2}{(1 - \nu_\lambda/n)^2}$$

where $\nu_\lambda = \mathrm{tr}(\mathbf{S}_\lambda)$ is the equivalent degrees of freedom.

Note that $\nu_\lambda = m + \nu_\lambda^1$ where $m = \mathrm{tr}(\mathbf{H}_w)$ and $\nu_\lambda^1 = \mathrm{tr}(\mathbf{S}_\lambda^1)$.

GCV has desirable asymptotic properties (e.g., see Li, 1987).

Reparameterization of PLS Problem

The PLS problem in Equation (1) can be rewritten as

$$\frac{1}{n}\|\mathbf{y}_w - \mathbf{X}_w\mathbf{b} - \mathbf{R}_w\mathbf{c}\|^2 + \lambda\mathbf{c}'\mathbf{\Psi}_w\mathbf{c} \tag{4}$$

where

- $\mathbf{R}_w = \sqrt{n}\mathbf{U}_w\mathbf{\Gamma}_w$ is the reparameterized contrast space matrix
- $\mathbf{U}_w\mathbf{D}_w\mathbf{V}_w'$ denotes the singular value decomposition of $\mathbf{P}_w$
- $\mathbf{\Gamma}_w\mathbf{\Psi}_w\mathbf{\Gamma}_w'$ denotes the eigen decomposition of $n\mathbf{D}_w^{-1}\mathbf{V}_w'\mathbf{Q}\mathbf{V}_w\mathbf{D}_w^{-1}$

Note that this diagonalizes and orthogonalizes the problem...

- $\mathbf{R}_w'\mathbf{R}_w = n\mathbf{I}$
- $\mathbf{\Psi}_w$ is diagonal
- $\mathbf{X}_w'\mathbf{R}_w = \mathbf{0}$

Separation of Reparameterized PLS Problem

Minimizing Equation (4) w.r.t. $(\mathbf{b}, \mathbf{c})$ is equivalent to solving

$$\min_{\mathbf{b}\in\mathbb{R}^m} \frac{1}{n}\|\mathbf{y}_w - \mathbf{X}_w\mathbf{b}\|^2 \quad \text{and} \quad \min_{\mathbf{c}\in\mathbb{R}^r} \frac{1}{n}\|\mathbf{y}_w - \mathbf{R}_w\mathbf{c}\|^2 + \lambda\mathbf{c}'\boldsymbol{\Psi}_w\mathbf{c} \tag{5}$$

and defining $\mathbf{g}_w = \frac{1}{n}\mathbf{R}_w'\mathbf{y}_w$ the optimal coefficients have the form

$$\hat{\mathbf{b}} = (\mathbf{X}_w'\mathbf{X}_w)^{-1}\mathbf{X}_w'\mathbf{y}_w \quad \text{and} \quad \hat{\mathbf{c}}_\lambda = (\mathbf{I} + \lambda\boldsymbol{\Psi}_w)^{-1}\mathbf{g}_w \tag{6}$$

The original coefficients can be recovered via a linear transformation

$$\begin{pmatrix} \hat{\boldsymbol{\beta}}_\lambda \\ \hat{\boldsymbol{\gamma}}_\lambda \end{pmatrix} = \begin{pmatrix} \mathbf{I} & -(\mathbf{X}_w'\mathbf{X}_w)^{-1}\mathbf{X}_w'\mathbf{Z}_w\mathbf{T}_w \\ \mathbf{0} & \mathbf{T}_w \end{pmatrix} \begin{pmatrix} \hat{\mathbf{b}} \\ \hat{\mathbf{c}}_\lambda \end{pmatrix}$$

where $\mathbf{T}_w = \sqrt{n}\mathbf{V}_w\mathbf{D}_w^{-1}\boldsymbol{\Gamma}_w$ (because $\mathbf{R}_w = \mathbf{P}_w\mathbf{T}_w$).

Smoothing and Selection via Elastic Net

Consider the Elastic Net (Zou & Hastie, 2005) version of Equation (5)

$$R_{\lambda,\alpha}(\mathbf{c}) = L(\mathbf{c}) + \lambda P_\alpha(\mathbf{c}) \tag{7}$$

where $L(\mathbf{c}) = \frac{1}{2n}\|\mathbf{y}_w - \mathbf{R}_w\mathbf{c}\|^2$ and

$$P_\alpha(\mathbf{c}) = \sum_{v=1}^{r} \psi_v \left[\tfrac{1}{2}(1-\alpha)c_v^2 + \alpha|c_v|\right]$$

is the (weighted) Elastic Net penalty.

- ψ_v are the diagonal elements of $\mathbf{\Psi}_w$
- $\mathbf{c} = (c_1, \ldots, c_r)'$ are reparameterized coefficients

Note $\alpha = 0$ produces the classic (ridge) solution in Equation (6).

Elastic Net Coefficient Estimates

The derivatives of Equation (7) imply the optimal coefficients satisfy

$$\hat{c}_v = \frac{S(g_v, \lambda\alpha\psi_v)}{1 + \lambda(1-\alpha)\psi_v} \tag{8}$$

where

- $g_v = \frac{1}{n}\mathbf{r}_v'\mathbf{y}_w$ is the v-th element of $\mathbf{g}_w = (g_1, \ldots, g_r)'$
- $S(a, b) = \text{sign}(a)(|a| - b)_+$ is the soft-thresholding operator

Letting $\hat{\mathbf{c}}_{\lambda,\alpha} = (\hat{c}_1, \ldots, \hat{c}_r)'$, we can write

$$\hat{\mathbf{c}}_{\lambda,\alpha} = [\mathbf{I} + \lambda(1-\alpha)\boldsymbol{\Psi}_w]^{-1}\boldsymbol{\Delta}_{\mathcal{A}}(\mathbf{I} - \lambda\alpha\boldsymbol{\Psi}_w\mathbf{G})\mathbf{g}_w$$

where

- $\mathbf{G} = \text{diag}(1/|g_1|, \ldots, 1/|g_r|)$ depends on the response vector
- $\boldsymbol{\Delta}_{\mathcal{A}}$ defines active set, i.e., $\delta_v^{\mathcal{A}} = 1$ if $|g_v| > \lambda\alpha\psi_v$ and $\delta_v^{\mathcal{A}} = 0$ o/w

Elastic Net Fitted Values

The elastic net fitted values can be written as

$$\begin{aligned}\hat{\mathbf{y}}_{\lambda,\alpha} &= \mathbf{X}\hat{\mathbf{b}} + \mathbf{R}\hat{\mathbf{c}}_{\lambda,\alpha} \\ &= \mathbf{W}^{-1/2}(\mathbf{S}_{\lambda,\alpha} - \mathbf{E}_{\lambda,\alpha})\mathbf{y}_w\end{aligned}$$

where $\mathbf{R} = \mathbf{W}^{-1/2}\mathbf{R}_w$.

Note that the smoothing and shrinkage matrices have the form

- $\mathbf{S}_{\lambda,\alpha} = \begin{bmatrix}\mathbf{X}_w & \mathbf{R}_w\boldsymbol{\Delta}_{\mathcal{A}}\end{bmatrix} \begin{bmatrix}\mathbf{X}_w'\mathbf{X}_w & \mathbf{0} \\ \mathbf{0} & n\mathbf{I} + n\lambda(1-\alpha)\boldsymbol{\Psi}_w\end{bmatrix}^{-1} \begin{bmatrix}\mathbf{X}_w' \\ \boldsymbol{\Delta}_{\mathcal{A}}\mathbf{R}_w'\end{bmatrix}$
- $\mathbf{E}_{\lambda,\alpha} = \lambda\alpha\mathbf{R}_w\boldsymbol{\Delta}_{\mathcal{A}}\left[n\mathbf{I} + n\lambda(1-\alpha)\boldsymbol{\Psi}_w\right]^{-1}\boldsymbol{\Psi}_w\mathbf{G}\boldsymbol{\Delta}_{\mathcal{A}}\mathbf{R}_w'$

GCV for Elastic Net Tuning

The GCV criterion can be written as

$$\mathrm{GCV}(\lambda, \alpha) = \frac{\frac{1}{n}\|(\mathbf{I} - \mathbf{S}_{\lambda,\alpha} + \mathbf{E}_{\lambda,\alpha})\mathbf{y}_w\|^2}{(1 - \nu_{\lambda,\alpha}/n)^2}. \tag{9}$$

where $\nu_{\lambda,\alpha}$ is the degrees of freedom of the Elastic Net solution.

The $\nu_{\lambda,\alpha}$ parameter can be estimated as the trace of $\mathbf{S}_{\lambda,\alpha}$ such as

$$\hat{\nu}_{\lambda,\alpha} = \mathrm{tr}(\mathbf{S}_{\lambda,\alpha}) = m + \sum_{v=1}^{r} \frac{\delta_v^{\mathcal{A}}}{1 + \lambda(1-\alpha)\psi_v}$$

and the numerator of the GCV criterion can be written as

$$\frac{1}{n}\|(\mathbf{I} - \mathbf{S}_{\lambda,\alpha} + \mathbf{E}_{\lambda,\alpha})\mathbf{y}_w\|^2 = \frac{1}{n}\|\mathbf{y}_w - \mathbf{X}_w\hat{\mathbf{b}}\|^2 - 2\mathbf{g}_w'\hat{\mathbf{c}}_{\lambda,\alpha} + \|\hat{\mathbf{c}}_{\lambda,\alpha}\|^2$$

GCV is Non-Convex Function of λ for $\alpha > 0$

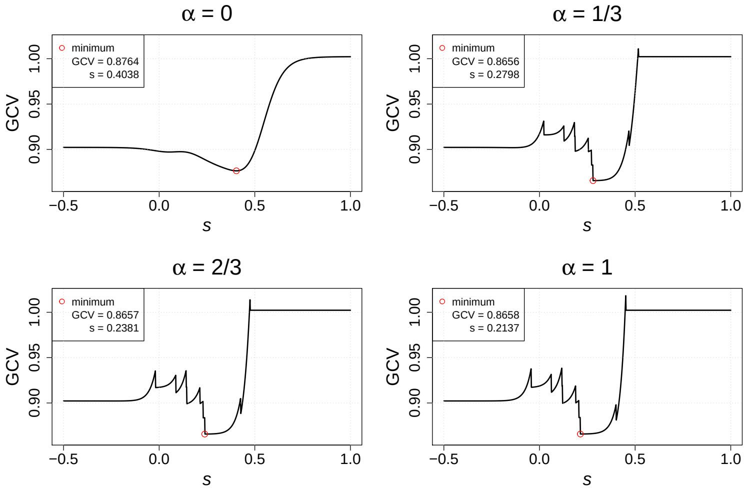

Figure 1: GCV as a function of $s = 1 + \log_{256}(\lambda)/3$ for different values of α.

Non-Convex Optimization Approach

Use the "fix α and tune λ" method to minimize GCV:

1. Choose sequence of α values (e.g., $\alpha = 0, 0.1, \ldots, 0.9, 1$)
2. For each α, find λ that minimizes $\mathrm{GCV}(\lambda, \alpha)$

Let $\lambda_{(v)}^{\alpha}$ denote v-th order statistic of $\{\lambda_v^{\alpha}\}_{v=1}^{r}$ where $\lambda_v^{\alpha} = |g_v|/(\alpha\psi_v)$

- $\lambda_{(1)}^{\alpha}, \ldots, \lambda_{(r)}^{\alpha}$ defines a solution path
- v coefficients in $\mathbf{c}$ are constrained to zero using $\lambda_{(v)}^{\alpha}$

If the minimum of the GCV occurs for $\lambda_{(1)}^{\alpha}$, GCV is convex below $\lambda_{(1)}^{\alpha}$

- Can use standard root finding algorithms to find minimum
- The `optimize` function in R can be used for this purpose

Extensions for Multiple Nonparametric Regression

Consider a multiple nonparametric regression model of the form

$$y_i = \eta(\mathbf{x}_i) + \epsilon_i$$

where

- $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})' \in \mathcal{X}$ is a $p > 1$ dimensional predictor
- $\mathcal{X} = \prod_{j=1}^{p} \mathcal{X}_j$ denotes some product domain
- $\mathcal{X}_j$ is the domain of x_{ij}

Can apply proposed ideas to tensor product design and penalty

- GAM: $\mathbf{X} = [\mathbf{X}_1, \ldots, \mathbf{X}_p]$ and $\mathbf{Z} = [\mathbf{Z}_1, \ldots, \mathbf{Z}_p]$
- SSANOVA: $\mathbf{X} = [\mathbf{X}_1, \ldots, \mathbf{X}_p]$ and $\mathbf{Z} = \sum_{j=1}^{p} \theta_j \mathbf{Z}_j$

Extensions for Generalized Nonparametric Regression

Consider a generalized nonparametric regression model of the form

$$g(\mu_i) = \eta(\mathbf{x}_i)$$

where

- $g(\cdot)$ is a known and invertible link function
- μ_i denotes the conditional mean of y_i given $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})' \in \mathcal{X}$
- y_i follows an exponential family distribution

Can apply proposed ideas to generalized NP regression models but…

- Need iteratively reweighted least squares (IRLS) for estimation
- Use **glmnet** (Friedman et al., 2010) to compute solution path

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Simulation A: Design & Analyses

Fully crossed simulation that manipulates two factors:

- exponential family distribution: Gaussian, Binomial, Poisson
- number of knots: $r \in \{10, 20, 30\}$

$\eta(x_i) = \sin(2\pi x_i)$ with $x_i = \frac{i-1}{n-1}$ for $i = 1, \ldots, n$ with $n = 100$

Compare kesso to `gam()` function from **mgcv** package (Wood, 2020)

- kesso implemented in **npnet** package from Helwig (2020)
- Uses **npreg** package (Helwig, 2021) for RK function evaluation

1000 replications for each combination of data generating conditions

Simulation A: Results

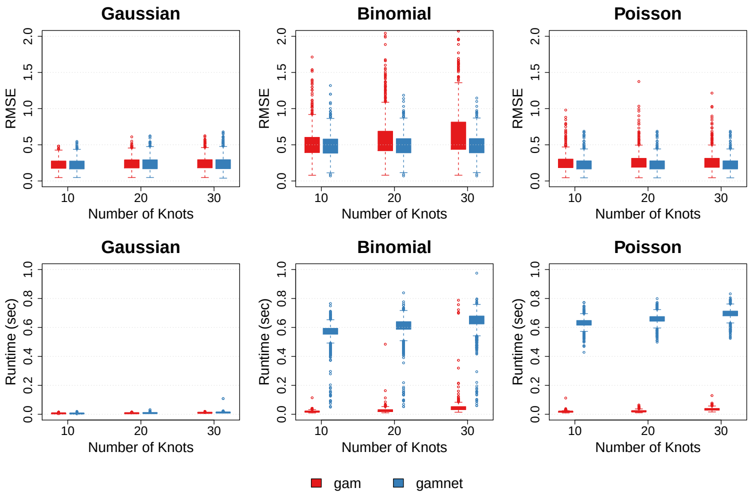

Figure 2: Root mean squared errors (top) and runtimes (bottom) for Simulation A.

Simulation A: Conclusions

For Gaussian response, there is no clear (dis)advantage of kesso.

- kesso and GAM performed similarly for all numbers of knots

For non-Gaussian response, kesso produced smaller RMSE than GAM

- Binomial: advantage of kesso increased as r increased
- Poisson: advantage of kesso consistent across r

Take-home: for single predictor kesso...

- can result in better function recovery
- is more computationally costly than GAM

Simulation B: Design & Analyses

Fully crossed simulation that manipulates two factors:

- data generating function: 4 functions (see next slide)
- number of observations: $n \in \{500, 1000, 2000\}$

Fit interaction model to all four functions (misspecified for A and C)

Compare kesso to two popular methods for smoothing:

- GAM using `gam()` from **mgcv** package (Wood, 2020)
- SSANOVA using `bigssp()` from `bigsplines` (Helwig, 2018)

1000 replications for each combination of data generating conditions

Simulation B: Data Generating Functions

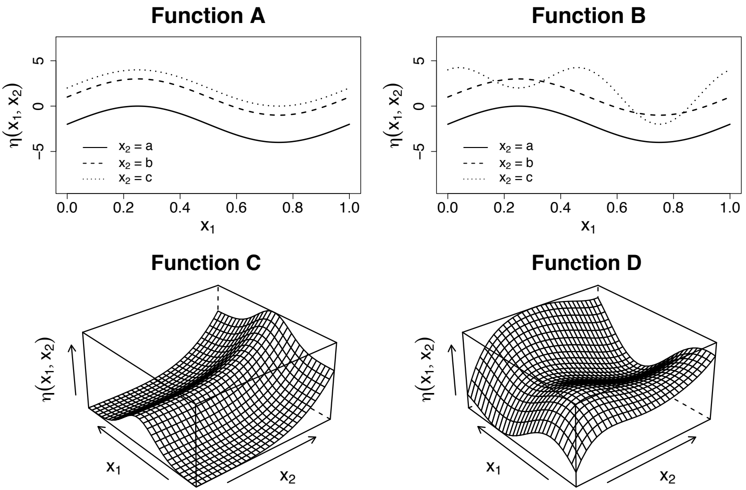

Figure 3: Data generating functions used in Simulation B.

Simulation B: Results

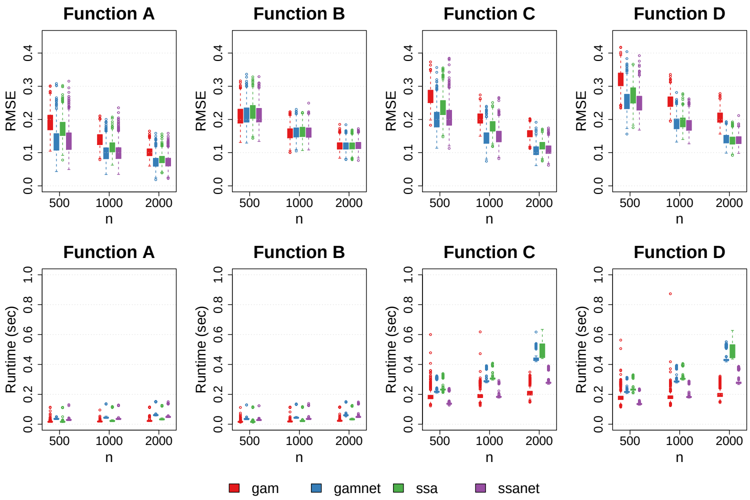

Figure 4: Root mean squared errors (top) and runtimes (bottom) for Simulation B.

Simulation B: Conclusions

kesso tends to outperform GAM and SSANOVA approaches

- Noteworthy reduction in RMSE for misspecified models (A and C)
- Reductions in RMSE for small samples (when correctly specified)

Comparing the run times[1] of the various methods:

- `gam` was faster than `gamnet` for all functions
- `ssa` was faster than `ssanet` for A and B but not C and D
- all methods take 0.5 seconds or less

Take home: kesso has advantages for misspecification and/or small n

[1] Hard to compare because **npnet** and **npreg** are pure R implementations

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Purpose of Real Data Examples

Using four real datasets, compare predictive performance of kesso to **glmnet** (Friedman et al., 2010), which is often used for elastic net.

For each dataset (see next slide), prediction performance is assessed by

1. splitting data into training and testing sets (70/30)
2. tuning (λ, α) on the training set[2]
3. evaluate RMSE on the testing set
4. repeat steps 1-3 for 100 random splits

Also recorded the amount of tuning time for kesso and **glmnet**

[2] Use 10-fold CV for **glmnet** and GCV for kesso

Overview of Four Datasets

Datasets from UCI Machine Learning Repository (Dua & Graff, 2019)

1. **Auto MPG**: $n = 392$ and $p = 7$. GOAL: Predict a car's miles per gallon (MPG) from the car's manufacturing features. Original source: Quinlan (1993).
2. **Bike Sharing**: $n = 17379$ and $p = 11$. GOAL: Predict (the log of) the number of rented bikes from various environmental factors. Original source: Fanaee-T & Gama (2013).
3. **Wine Quality**: $n = 1599$ and $p = 11$. GOAL: Predict the quality rating of red wine from a variety of chemical features. Original source: Cortez et al. (2009).
4. **Student Performance**: $n = 395$ and $p = 32$. GOAL: Predict the student's final exam grade from a variety of factors. Original source: Cortez & Silva (2008).

Prediction RMSE and Tuning Cost

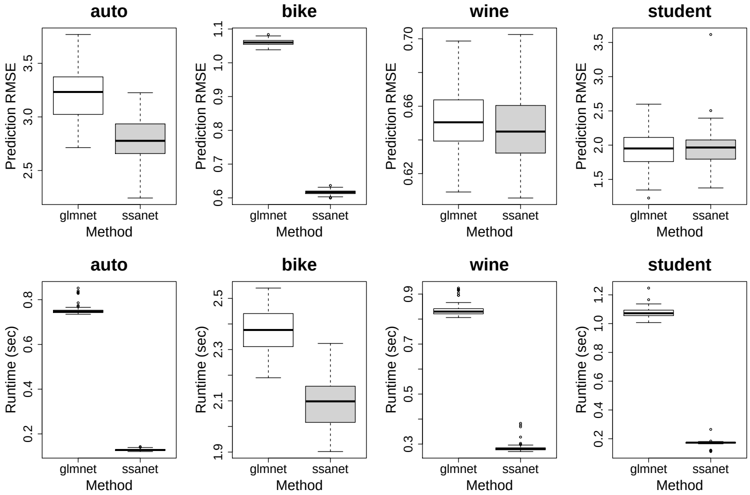

Figure 5: Prediction RMSE (top) and the runtime (bottom) for each real data example.

Variable Importances for Auto MPG Data

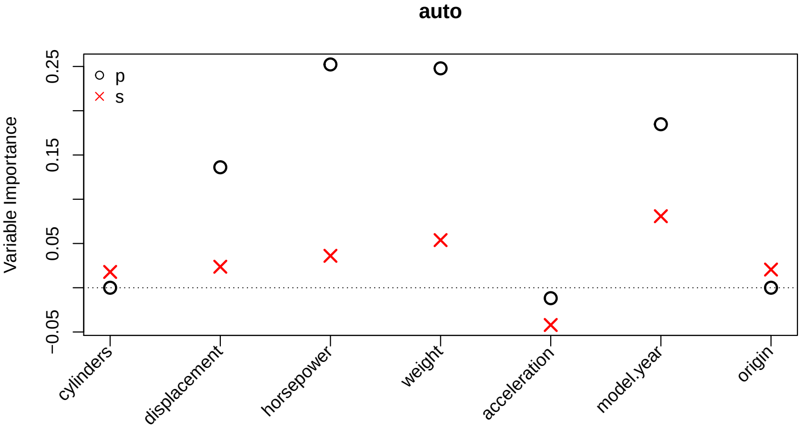

Figure 6: Variable importance indices for parametric (p) and smooth (s) effects.

Variable Importances for Bike Sharing Data

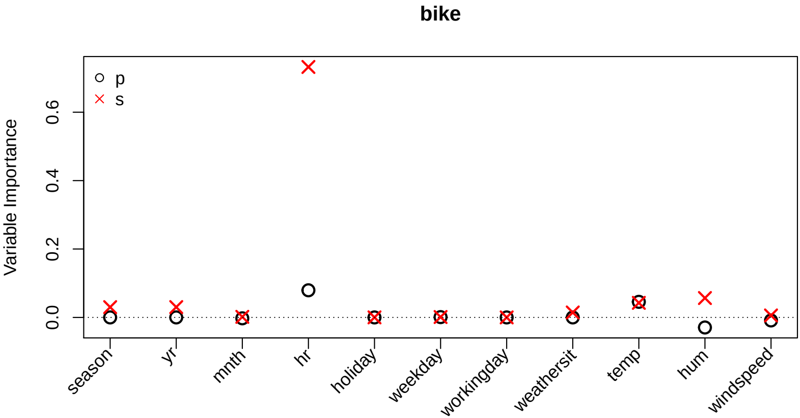

Figure 7: Variable importance indices for parametric (p) and smooth (s) effects.

Variable Importances for Wine Data

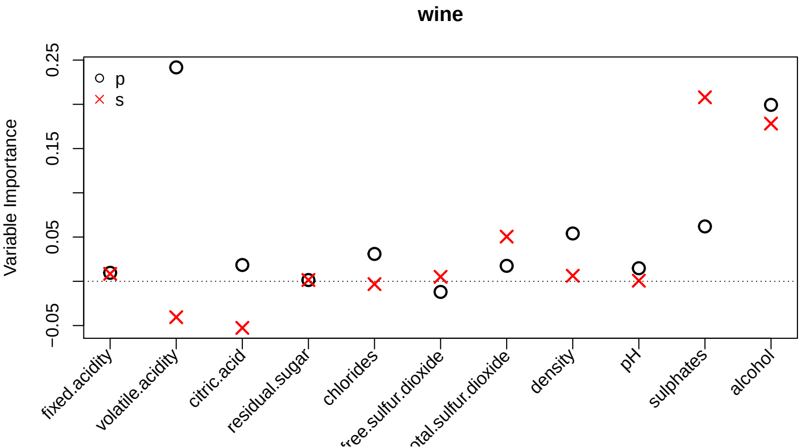

Figure 8: Variable importance indices for parametric (p) and smooth (s) effects.

Variable Importances for Student Data

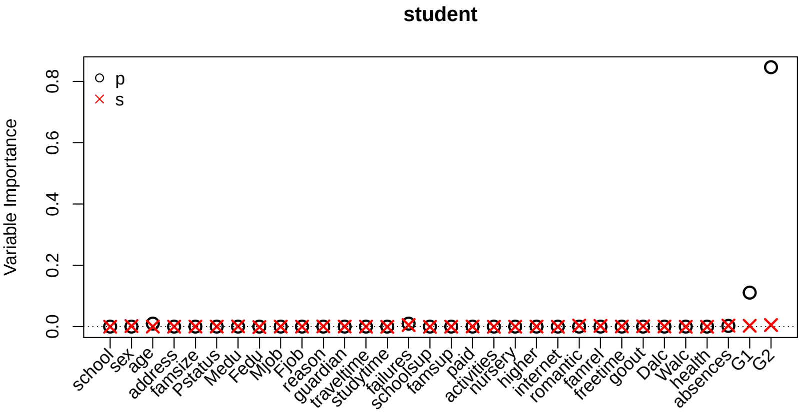

Figure 9: Variable importance indices for parametric (p) and smooth (s) effects.

Real Data Conclusions

For 3/4 datasets, kesso produces noteworthy prediction gains

- Important non-linear effects (e.g., year for auto & time for bike)
- Student dataset has relatively linear (conditional) effects

For all 4 datasets, GCV tuning was more efficient than k-fold CV

- For smaller n, kesso GCV tuning was substantially more efficient
- For larger n, computational gains are more modest

<u>Take home</u>: kesso outperforms popular GLMNET approach

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Take Home Messages

Proposed kesso can find truth within noisy data

- Performs as well as or better than GAM and SSANOVA
- Has advantages under misspecification and small n

Proposed kesso can predict well for real data

- Noteworthy prediction gains by allowing for non-linear effects
- GCV tuning is practical and computationally advantageous

Should kesso be the new default smoothing method?

- GAM and SSANOVA are special cases of kesso ($\alpha = 0$)
- Benefit in selecting eigenvectors of tensor product kernels ($\alpha > 0$)

Room for Improvement

Computational bottleneck is SVD of $\mathbf{P}_w = (\mathbf{I} - \mathbf{H}_w)\mathbf{Z}_w$

- Requires $O(nr^2)$ flops to compute SVD
- Can be costly or infeasible for large n and/or large r

Eigen decomposition of $n\mathbf{D}_w^{-1}\mathbf{V}_w'\mathbf{Q}\mathbf{V}_w\mathbf{D}_w^{-1}$ can be costly for large r

- Unclear how large r needs to be for real data applications
- Advisable to try multiple different values of r for real data

Optimal solution path for generalized models is open problem

- Gaussian responses permit efficient solution path via $\{\lambda_v\}_{v=1}^r$
- Non-Gaussian responses lack simple closed-form solution path

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Overview of Reparameterization

Getting from Equation (1) to Equation (4) involves three steps:

1. Orthogonalize null and contrast spaces
2. Reparameterize contrast space
3. Diagonalize smoothness penalty

In the following slides, each of these three ideas will be elaborated to show the equivalence between Equation (1) and Equation (4).

- Each idea builds off of the previous idea to create Equation (4)
- Can think of it as three reparameterizations combined into one

Step 1: Subspace Orthogonalization

Consider the reparameterized PLS problem

$$\frac{1}{n}\|\mathbf{y}_w - \mathbf{X}_w\mathbf{b} - \mathbf{P}_w\boldsymbol{\gamma}\|^2 + \lambda\boldsymbol{\gamma}'\mathbf{Q}\boldsymbol{\gamma} \tag{10}$$

which is Equation (1) with $\mathbf{Z}_w$ replaced by $\mathbf{P}_w$.

The optimal coefficients satisfy

$$\hat{\mathbf{b}} = (\mathbf{X}_w'\mathbf{X}_w)^{-1}\mathbf{X}_w'\mathbf{y}_w \qquad \text{and} \qquad \hat{\boldsymbol{\gamma}}_\lambda = (\mathbf{P}_w'\mathbf{P}_w + n\lambda\mathbf{Q})^{-1}\mathbf{P}_w'\mathbf{y}_w$$

where $\hat{\boldsymbol{\gamma}}_\lambda$ is the same estimate as denoted in Equation (2).

Original null space coefficients are $\hat{\boldsymbol{\beta}}_\lambda = \hat{\mathbf{b}} - (\mathbf{X}_w'\mathbf{X}_w)^{-1}\mathbf{X}_w'\mathbf{Z}_w\hat{\boldsymbol{\gamma}}_\lambda$

Step 2: Contrast Space Reparameterization

Consider a reparameterization of Equation (10) of the form

$$\frac{1}{n}\|\mathbf{y}_w - \mathbf{X}_w\mathbf{b} - \mathbf{U}_w\boldsymbol{\xi}\|^2 + \lambda\boldsymbol{\xi}'\tilde{\mathbf{Q}}\boldsymbol{\xi} \tag{11}$$

where $\boldsymbol{\xi} = \mathbf{D}_w\mathbf{V}_w'\boldsymbol{\gamma}$ and $\tilde{\mathbf{Q}} = \mathbf{D}_w^{-1}\mathbf{V}_w'\mathbf{Q}\mathbf{V}_w\mathbf{D}_w^{-1}$.

- Note that $\mathbf{U}_w\boldsymbol{\xi} = \mathbf{P}_w\boldsymbol{\gamma}$ and $\boldsymbol{\xi}'\tilde{\mathbf{Q}}\boldsymbol{\xi} = \boldsymbol{\gamma}'\mathbf{Q}\boldsymbol{\gamma}$

The optimal coefficients satisfy

$$\hat{\mathbf{b}} = (\mathbf{X}_w'\mathbf{X}_w)^{-1}\mathbf{X}_w'\mathbf{y}_w \qquad \text{and} \qquad \hat{\boldsymbol{\xi}}_\lambda = (\mathbf{I} + n\lambda\tilde{\mathbf{Q}})^{-1}\mathbf{U}_w'\mathbf{y}_w$$

given that $\mathbf{X}_w'\mathbf{U}_w = \mathbf{0}$ and $\mathbf{U}_w'\mathbf{U}_w = \mathbf{I}$.

Original contrast space coefficients are $\hat{\boldsymbol{\gamma}}_\lambda = \mathbf{V}_w\mathbf{D}_w^{-1}\hat{\boldsymbol{\xi}}_\lambda$

Step 3: Penalty Diagonalization

Consider a reparameterization of Equation (11) of the form

$$\frac{1}{n}\|\mathbf{y}_w - \mathbf{X}_w\mathbf{b} - \mathbf{R}_w\mathbf{c}\|^2 + \lambda\mathbf{c}'\boldsymbol{\Psi}_w\mathbf{c} \tag{12}$$

where $\mathbf{R}_w = n^{1/2}\mathbf{U}_w\boldsymbol{\Gamma}_w$ and $\mathbf{c} = n^{-1/2}\boldsymbol{\Gamma}_w'\boldsymbol{\xi}$.

- Note that $\mathbf{R}_w\mathbf{c} = \mathbf{U}_w\boldsymbol{\xi}$ and $\mathbf{c}'\boldsymbol{\Psi}_w\mathbf{c} = \boldsymbol{\xi}'\tilde{\mathbf{Q}}\boldsymbol{\xi}$

The optimal coefficients satisfy

$$\hat{\mathbf{b}} = (\mathbf{X}_w'\mathbf{X}_w)^{-1}\mathbf{X}_w'\mathbf{y}_w \qquad \text{and} \qquad \hat{\mathbf{c}}_\lambda = (n\mathbf{I} + n\lambda\boldsymbol{\Psi}_w)^{-1}\mathbf{R}_w'\mathbf{y}_w$$

given that $\mathbf{X}_w'\mathbf{R}_w = \mathbf{0}$ and $\mathbf{R}_w'\mathbf{R}_w = n\mathbf{I}$.

Reparameterized contrast space coefficient are $\hat{\boldsymbol{\xi}}_\lambda = n^{1/2}\boldsymbol{\Gamma}_w\hat{\mathbf{c}}_\lambda$

Elastic Net Problem Revisited

The elastic net version of the problem has the form

$$R_{\lambda,\alpha}(\mathbf{c}) = L(\mathbf{c}) + \lambda P_\alpha(\mathbf{c})$$

where

- $L(\mathbf{c}) = \frac{1}{2n}\|\mathbf{y}_w - \mathbf{R}_w\mathbf{c}\|^2$
- $P_\alpha(\mathbf{c}) = \sum_{v=1}^r \psi_v \left[\frac{1}{2}(1-\alpha)c_v^2 + \alpha|c_v|\right]$

ψ_v are the diagonal elements of $\mathbf{\Psi}_w$

c_v is the v-th element of $\mathbf{c} = (c_1, \ldots, c_r)'$.

Elastic Net Problem Derivatives

The derivative of the unpenalized portion of the loss function is

$$\frac{\partial L(\mathbf{c})}{\partial c_v} = c_v - \frac{1}{n}\mathbf{r}_v'\mathbf{y}_w$$

given that $\|\mathbf{r}_v\|^2 = n$ and $\mathbf{r}_u'\mathbf{r}_v = 0$ for $u \neq v$.

The derivative of the penalty has the form

$$\frac{\partial \lambda P_\alpha(\mathbf{c})}{\partial c_v} = \begin{cases} \lambda\psi_v\left[(1-\alpha)c_v + \alpha\right] & \text{if } c_v > 0 \\ \lambda\psi_v\left[(1-\alpha)c_v - \alpha\right] & \text{if } c_v < 0 \end{cases}$$

and is undefined if $c_v = 0$.

$\hat{c}_v = \frac{S(g_v, \lambda\alpha\psi_v)}{1+\lambda(1-\alpha)\psi_v}$ where $g_v = \frac{1}{n}\mathbf{r}_v'\mathbf{y}_w$, which is the result in Equation (8)